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Predicting absorption spectra of silver-ligand complexes

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Abstract

The detailed structures of most of ligand-stabilized metal nanoclusters (NCs) remain unknown due to the absence of crystal structure data for them. In such a situation, quantum-chemical modeling is of particular interest. We compared the performance of different theoretical methods of geometry optimization and absorption spectra calculation for silver-thiolate complexes. We showed that the absorption spectra calculated with the ADC(2) method were consistent with the spectra obtained with CC2 method. Three DFT functionals (B3LYP, CAM-B3LYP, and M06-2X) failed to reproduce the CC2 absorption spectra of the silver-thiolate complexes.

KEYWORDS
ADC(2), CAM-B3LYP, CC2, M062X, silver nanoclusters, silver-thiolate complexes, TDDFT

1 | INTRODUCTION

Ligand protected metal nanoclusters (NCs) have gained much attention in the last decade due to their unique physicochemical properties. These metal NCs typically measure less than 2 nm.[1,2] Electronic, chemical, and optical properties of NCs differ from both bulk and atomic scale materials. Optically active NCs may possess strong visible or near infrared luminescence,[3] which can be used in various applications, such as bioimaging, biosensing, and optical detection of biomolecules.[4] Among the great variety of NCs, gold (Au) NCs are the most extensively investigated due to their chemical stability.[5] However, fluorescent silver (Ag) clusters are the brightest, exhibiting high quantum yield and large absorption cross-section.[6]

Various techniques have been developed for the synthesis of stable, water-soluble silver NCs by using dendrimers,[7] polymers,[8] polypeptides,[9,10] thiols,[11] DNA,[12,13] thiolated DNA,[14] and other ligands to stabilize silver ions.[15] The influence of various thiolates on binding energy with silver NCs has been studied in details.[16] The nature of the ligand seems to have a strong effect on fluorescent properties of noble metal NCs.[17] Noble metal NCs protected with thiolate ligands have been of interest because of their long-term stability, which allows one to use them as building blocks constructing assembled systems with novel and improved functions.[2] Silver-thiolate and silver-amino acid complexes are studied in many laboratories nowadays both theoretically and experimentally.[18–20] It is assumed that silver nanoclusters are stabilized by cysteine residues in protein matrices.[21,22] While the composition of the ligand-protected NCs can be analyzed using mass-spectroscopy technique,[23] their detailed structures remain unknown in many cases. In this respect, quantum-chemical modeling of silver-ligand complexes is of particular interest.[24–32] Gell and coworkers investigated thiolate-protected silver NCs that contained zero, two, or four confined electrons in the cluster core.[33] They found that the number of confined electrons determine the spectroscopic patterns. For systems with two or four confined electrons, strong absorption and fluorescence in the visible or even IR energy regime can be found. Therefore, in our study we investigated small silver-thiolate NCs with two confined electrons in the cluster core.

It should be noted that most theoretical studies used the density functional theory (DFT) approach, namely time-dependent (TDDFT) approach, when calculating optical transitions.[24–31,34,35] Meanwhile, TDDFT can significantly underestimate energies of charge-transfer (CT) states,[36–38] which makes such approach inappropriate for accurate modeling of the electronic spectra for organometallic clusters in the case of electron transfer. Although long-range corrected, or range-separated hybrid functionals (eg, CAM-B3LYP) may be good in this case, several aspects should be taken into account: for example, the choice of empirical parameters, and size-consistency violation.[39,40] The validity of the

TDDFT approach should be verified by an appropriate experimental or theoretical reference. At the same time, application of pure ab initio methods is limited to medium-size systems. In this respect, finding a cost-effective strategy for real thiolate-silver complexes is such an important and challenging task.

In this article, we studied the performance of different theoretical methods predicting the geometry of silver-thiolate complexes. Then, we compared the efficiency of different methods computing the absorption spectra of silver-thiolate complexes.

2 | METHODS

The geometries of all complexes were optimized using second-order Moller-Plesset perturbation theory (MP2) realized in Orca v.3.0 program package[41] and with resolution-of-the-identity second-order approximate coupled-cluster singles and doubles (RI-CC2) method[42] realized in Turbomole v.7.2 program package.[43] RI-CC2 and MP2 optimization methods used the same split basis set of def2-SVP for hydrogen atoms, def2-TZVP for carbon, oxygen, and sulfur atoms and def2-TZVPP for silver- def2-TZVP[P].[44] The electronic absorption spectra of the aforementioned complexes were calculated in Turbomole with second order algebraic-diagrammatic construction ADC(2) method.[45] Both RI-CC2 and ADC(2) method used the same split basis set of def2-SVP for hydrogen atoms, def2-QZVP for carbon, oxygen, and sulfur atoms and def2-QZVPP for silver - def2-QZVP(P).[44] Geometry optimization and calculation of electronic absorption spectra using M062X,[46] CAM-B3LYP,[47] and B3LYP[48,49] functionals were done in Gaussian v.09.[50]

3 | RESULTS AND DISCUSSION

3.1 | Comparison of geometry optimization methods

We studied the performance of different methods for geometry optimization. To do so, we compared CC2, MP2, and DFT geometries of $[Ag_2Th]^{-1}$, $[Ag_2Th_2]^{-2}$, $[Ag_3Th_2]^{-1}$, $[Ag_3Th_2A]^{-1}$, $[Ag_4Th_3]^{-1}$, and $[Ag_5Th_4]^{-1}$ (where Th stands for thiolate and A stands for formaldehyde) complexes.

Thus, the values of bond lengths and valence angles for $[Ag_4Th_3]^{-1}$ complex are presented in Figure 1 and in Table S1. The geometry parameters of MP2 and CC2 optimization do not differ significantly with each other, except for the Ag1-S2-Ag2 triangle. The bond lengths of the triangle are 2.35 Å for MP2 and 3.14 Å for CC2, while the valence angle is equal to 82.9° for MP2 and 57.3° for CC2. The geometrical parameters of $[Ag_4Th_3]^{-1}$ complex are also presented in Table S1. DFT functionals B3LYP, CAM-B3LYP, and M06-2X nicely reproduce the bond lengths and valence angles of $[Ag_4Th_3]^{-1}$ complex obtained with CC2 method, except for the Ag1-S2-Ag2 triangle. However, geometrical parameters of Ag1-S2-Ag2 triangle obtained using MP2 are reproduced quite correctly by DFT. Nevertheless, DFT fails to reproduce the dihedral angles of $[Ag_4Th_3]^{-1}$ (Table S1): for example, C1-S1-Ag1-S2 dihedral angle is equal to −164.1°, −48.9°, −64.4°, and − 129.6° for CC2, B3LYP, CAM-B3LYP, and M06-2X, respectively.

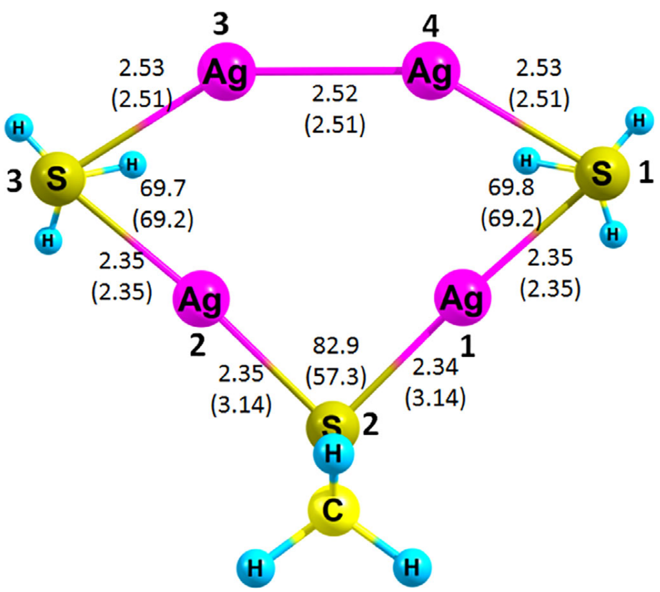

FIGURE 1 Geometry of $[Ag_4Th_3]^{-1}$ silver-thiolate complex optimized using MP2 method. The parameters of CC2 optimization are presented in parentheses. In both cases same split basis set def2-TZVP(P) was used

We also compared the performance of different methods on $[Ag_3Th_2]^{-1}$ geometry (Figure 2). The difference in geometrical parameters of MP2 and CC2 methods is less pronounced than in the case of $[Ag_4Th_3]^{-1}$ complex. The values of valence angles do not differ significantly. For example, the Ag1-S2-Ag2 angle is equal to 75.4° for CC2 and is equal to 77.4° (Table 1) when computed with MP2. All three functionals (B3LYP, CAM-B3LYP, and M06-2X) reproduce bond distances quiet well: the biggest difference with CC2 is observed for Ag2-Ag3 bond; it is equal to 2.52 nm, 2.63 nm, 2.61 nm, and 2.73 nm for CC2, B3LYP, CAM-B3LYP, and M06-2X, respectively. Things are getting worse when we regard valence angles, because all three functionals fail to reproduce the values of Ag1-S2-Ag2 angle: it is equal to 75.4°, 103.9°, 103.5°, and 83.7° for CC2, B3LYP, CAM-B3LYP, and M06-2X, respectively. Moreover, all three functionals fail to reproduce the values of dihedral angles of $[Ag_3Th_2]^{-1}$ silver-thiolate complex (Table 1).

The rest of the complexes ($[Ag_2Th]^{-1}$, $[Ag_2Th_2]^{-2}$, $[Ag_3Th_2A]^{-1}$, and $[Ag_5Th_4]^{-1}$) demonstrate the full agreement between MP2 and CC2 methods (Figure S1). The detailed data about $[Ag_2Th]^{-1}$ geometry is given in Table S2. The geometrical parameters obtained with five different

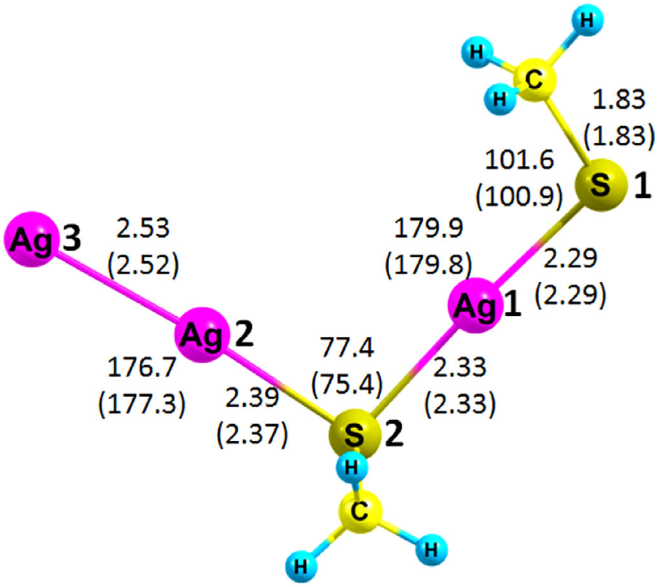

FIGURE 2 Geometry of $[Ag_3Th_2]^{-1}$ silver-thiolate complex optimized using MP2 method. The parameters of CC2 optimization are presented in parentheses. In both cases same basis set def2-TZVP(P) was used

TABLE 1 Geometrical parameters of $[Ag_3Th_2]^{-1}$ silver-thiolate complex optimized using MP2, CC2, and DFT

Bond lengths, Å	C1-S1	S1-Ag1	Ag1-S2	S2-C2	S2-Ag2	Ag2-Ag3
MP2	1.829	2.289	2.327	1.830	2.386	2.530
CC2	1.832	2.286	2.327	1.833	2.372	2.517
B3LYP	1.837	2.366	2.412	1.838	2.475	2.632
CAM-B3LYP	1.823	2.347	2.387	1.824	2.446	2.607
M06-2X	1.826	2.407	2.476	1.828	2.549	2.726

Valence angles, °	C1-S1-Ag1	S1-Ag1-S2	Ag1-S2-Ag2	S2-Ag2-Ag3	Ag2-S2-C2
MP2	101.6	179.9	77.4	176.7	102.2
CC2	100.9	179.8	75.4	177.3	102.3
B3LYP	104.9	175.8	103.9	176.0	103.6
CAM-B3LYP	104.9	176.5	103.5	176.4	103.6
M06-2X	102.7	177.6	83.7	180.0	101.8

Dihedral angles, °	Ag1-S2-Ag2-Ag3	Ag3-Ag2-S2-C2	S1-Ag1-S2-C2
MP2	−136.3	−34.6	−149.6
CC2	−136.5	−35.3	−139.2
B3LYP	−172.8	−63.5	86.9
CAM-B3LYP	−167.5	−58.2	84.7
M06-2X	−143.0	−40.9	75.6

Note: In all cases same def2-TZVP(P) basis set was used.

methods are in agreement with each other. The structure data about $[Ag_2Th_2]^{-2}$ geometry is given in Table S3. DFT correctly reproduces bond distance and valence angles, but two of three functionals (B3LYP and M06-2X) fail to estimate C1-S1-Ag1-Ag2 dihedral angle: according to CC2, B3LYP, and M06-2X the dihedral is equal to −10.4°, 143.5°, and 14.8°, respectively. The detailed comparison of $[Ag_3Th_2A]^{-1}$ geometry is given in Table S4. The structures obtained with CC2, MP2, CAM-B3LYP, and M06-2X are consistent with each other. However, the B3LYP geometry totally differs. The formaldehyde is attached to S1 sulfur atom (Figure S2) instead of S2 as in geometries obtained with four other methods (Figure S1). The comparison of $[Ag_5Th_4]^{-1}$ geometry is presented in Table S5: all five geometries are consistent with each other.

Thus, we demonstrated that geometrical parameters of silver-thiolate complexes calculated with MP2 method are similar to the geometrical parameters obtained with CC2 method. That means MP2 method can be used along with CC2 to perform geometry optimization of silver-thiolate complexes and allows to reduce computational costs. However, geometries obtained with DFT differ from CC2 geometries more or less: in general, DFT tends to overestimate Ag-S and Ag-Ag bond distances and fails to reproduce certain dihedral angles.

3.2 | UV-Vis spectra benchmark

We compared the performance of different theoretical methods for calculating optical absorption spectra of silver-thiolate complexes. In order to find the best one, we compared the performance of different ab initio methods: equation of motion coupled cluster singles and doubles (EOM-CCSD), RI-CC2, and ADC(2). We also calculated spectra of the complexes at M06-2X and CAMB3LYP level of theory. We examined the performance of all aforementioned methods using a small silver-thiolate complex, namely $[Ag_2Th]^{-1}$ (Figure S1).

We found that the electronic transitions calculated with EOM-CCSD, RI-CC2, and ADC(2) methods are consistent with each other (Table S6) especially when def2-QZVP(P) basis set is used. Thus, the results of EOM-CCSD method with the aforementioned hybrid basis set were used as reference I. The first transition in the absorption spectrum of $[Ag_2Th]^{-1}$ has a wavelength of 402 nm with an oscillator strength equal to 0.6. The second reference calculation is RI-CC2 while the reference III is ADC(2) method with same basis set.

The usage of different basis sets contributes adjustments into the calculated absorption spectrum. The application of LANL effective core potential on silver gives a strong bathochromic shift as compared to the reference EOM-CCSD method. Surprisingly, EOM-CCSD with the aug-cc-pVTZ{C,S}/aug-cc-pVTZ-PP{Ag} basis set gives a strong long-wave shift even in comparison with aug-cc-pVDZ{C,S}/aug-cc-pVDZ-PP{Ag} (Table S6).

The most suitable approximation of DFT is M06-2X with def2-TZVP(P) basis set. It has a first transition of 401 nm and 0.65 oscillator strength, the second transition is equal 359 nm and oscillator strength (f_{osc}) equals 0.26 (which is in agreement with the second transition of 359 nm [0.27] of reference I), the third transition is equal 347 nm that is in full agreement with reference II (348 nm and f_{osc} = 0.18).

Next, we considered other silver-thiolate complexes $[Ag_nTh_{(n-1)}]^{-1}$, where Th stands for —SCH_3 ligand. Complexes with two confined electrons $[Ag_3Th_2]^{-1}$, $[Ag_4Th_3]^{-1}$and $[Ag_5Th_4]^{-1}$were investigated earlier at CAMB3LYP level of the theory.[33] We added several other silver-ligand complexes in this study: $[Ag_2Th]^{-1}$, $[Ag_2Th_2]^{-2}$, and $[Ag_2Th_2A]^{-1}$, which possess formaldehyde (A) residue. We calculated electronic spectra of the complexes using high-level ab initio methods that would eliminate uncertainties related to CT interactions that may not be accounted for properly in the TDDFT method. We focused on the first five transitions in the absorption spectra and calculated the corresponding oscillator strengths.

The results are presented in Table 2. All calculations were performed using RI-CC2 optimized geometries. The calculations made by the ADC (2) and RI-CC2 methods are consistent with each other, once again validating the calculation approach with the split basis set. However, the first excitation is systematically higher for 10-24 nm when computed with ADC(2) method; the average deviation from RI-CC2 is 0.07 eV.

The results are consistent in the case of $[Ag_2Th]^{-1}$ for both RI-CC2 and ADC(2). The first transition is located at 390-400 nm followed by a CT at approximately 360 nm. CAM-B3LYP was used to predict the absorption spectra of silver-thiolate complexes in the paper by Gell and coworkers.[33] Both CAM-B3LYP and M062X fail to reproduce this latter CT transition with zero oscillator strength. We plot electron density difference (EDD) maps in Figure 3. The lowest energy transition at 404 nm is formed as electron density redistribution produced by absorption to the lowest state that is represented through EDD between the involved electronic states calculated using ADC(2) (Figure 3).

Both RI-CC2 and ADC(2) simulate similar spectra for $[Ag_2Th_2]^{-2}$ with first transition located at 577 nm for ADC(2) and 580 nm for RI-CC2. The M062X functional, however, cannot reproduce properly this spectrum with the first transition red-shifted to 709 nm. The CAM-B3LYP functional reproduced first transition at 611 nm, which is very close to the values obtained using the ab initio methods. Absorption involves both Ag cluster and ligand parts according to delocalized EDD (Figure 3).

All methods nicely reproduce the absorption spectrum of $[Ag_3Th_2]^{-1}$, however, in the case of M062X the first transition demonstrates a slight bathochromic shift as compared to RI-CC2 and ADC(2): 416 nm (0.561) for M062X, 383 nm (0.462) for RI-CC2 and 395 nm (0.566) for ADC(2). The first absorption band is presented on Figure 3 because the lowest state is mostly excited through CT from Ag to formaldehyde. CAM-B3LYP reproduces correctly this first transition with a value of 387 nm; however, it slightly overestimates the oscillator strength of this transition (0.520). When formaldehyde (A) is added to this $[Ag_3Th_2]^{-1}$ complex, M062X fails to reproduce its absorption spectrum indicating that first transition at 500 nm has a 0.001 oscillator strength. The same is true for CAM-B3LYP method: the first transition at 442 nm has zero oscillator strength. However, the spectra of ADC(2) and RI-CC2 for $[Ag_3Th_2A]^{-1}$ are consistent with each other. If we omit the first transition in the DFT absorption

TABLE 2 First five excitation transitions (in nm; oscillator strengths—in parentheses) for silver-thiolate complexes according to RI-CC2, ADC(2), M062X, and CAM-B3LYP method with the def2-QZVP(P) basis set (RI-CC2/def2-TZVP(P) optimization)

Method	$[Ag_2Th]^{-1}$	$[Ag_2Th_2]^{-2}$	$[Ag_3Th_2]^{-1}$	$[Ag_3Th_2A]^{-1a}$	$[Ag_4Th_3]^{-1}$	$[Ag_5Th_4]^{-1}$
RI-CC2	394 (0.476)	580 (0.116)	383 (0.462)	413 (0.110)	585 (0.021)	463 (0.025)
	358 (0.000)	570 (0.115)	341 (0.129)	390 (0.031)	462 (0.162)	438 (0.123)
	352 (0.152)	547 (0.161)	326 (0.163)	385 (0.015)	457 (0.178)	402 (0.000)
	342 (0.182)	448 (0.010)	318 (0.011)	365 (0.002)	448 (0.037)	386 (0.004)
	329 (0.003)	436 (0.006)	313 (0.022)	357 (0.014)	387 (0.000)	381 (0.014)
ADC(2)	404 (0.582)	577 (0.205)	395 (0.566)	427 (0.095)	607 (0.032)	473 (0.037)
	362 (0.000)	567 (0.199)	348 (0.211)	400 (0.020)	475 (0.273)	448 (0.165)
	357 (0.256)	540 (0.185)	331 (0.285)	396 (0.016)	469 (0.218)	411 (0.000)
	346 (0.307)	446 (0.015)	323 (0.021)	376 (0.002)	456 (0.029)	392 (0.010)
	332 (0.003)	439 (0.010)	317 (0.010)	366 (0.019)	393 (0.003)	387 (0.004)
M06-2X	442 (0.501)	709 (0.123)	416 (0.561)	500 (0.001)	629 (0.040)	466 (0.061)
	410 (0.263)	700 (0.170)	395 (0.191)	409 (0.218)	486 (0.151)	423 (0.225)
	401 (0.279)	670 (0.192)	373 (0.281)	381 (0. 061)	485 (0.072)	389 (0.006)
	347 (0.030)	557 (0.017)	330 (0.090)	377 (0.031)	481 (0.254)	371 (0.047)
	342 (0.021)	536 (0.008)	318 (0.007)	360 (0.015)	425 (0.002)	347 (0.011)
CAM-B3LYP	409 (0.510)	611 (0.157)	387 (0.520)	442 (0.000)	524 (0.050)	416 (0.049)
	374 (0.254)	599 (0.153)	353 (0.213)	384 (0.008)	446 (0.260)	404 (0.185)
	367 (0.257)	561 (0.152)	340 (0.262)	380 (0.130)	421 (0.218)	343 (0.003)
	341 (0.021)	464 (0.014)	305 (0.007)	353 (0.026)	398 (0.015)	330 (0.000)
	316 (0.002)	451 (0.009)	295 (0.022)	351 (0.052)	363 (0.005)	327 (0.062)

[a]A = O=CH_2; Th = S-CH_3.

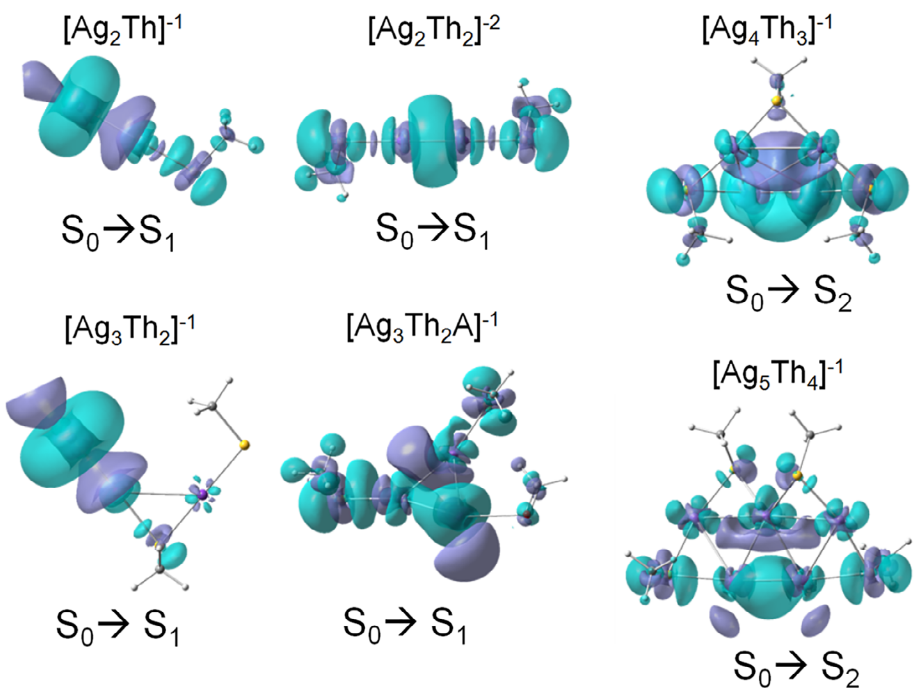

FIGURE 3 Electron density difference (EDD) for the most dominant transitions in the absorption spectra of silver-thiolates according to ADC (2) method (dark lilac areas illustrate positive CT directions with grid of 0.001 $e^{-}/bohr^3$ of EDD)

spectra of $[Ag_3Th_2A]^{-1}$, the dominant transition is predicted by M062X (409 nm) and CAM-B3LYP (380 nm) quite correctly as compared to RI-CC2 (413 nm).

The spectra of $[Ag_4Th_3]^{-1}$ simulated by RI-CC2 and ADC(2) are in agreement with each other. According to RI-CC2 the first transition is located at 585 nm, it is situated at 607 nm according to ADC(2). The main contribution to this transition comes from Ag with minor delocalization on the ligands (Figure 3). This first electronic transition is bathochromicly shifted to 629 nm according to M062X, while in the case of CAM-B3LYP functional the first transition is significantly blue-shifted to 524 nm. The RI-CC2 spectrum is dominated by the transition at 457 nm (0.178), this dominant transition is located at 469 nm (0.218) in ADC(2) spectrum. The bathochromic shift is observed in the absorption spectrum of M062X for this transition as compared with ADC(2) and RI-CC2: 481 nm (0.254). The CAM-B3LYP spectrum is dominated by the transition at 421 nm (0.218), which is hypsochromicly shifted as compared to other methods.

The first transition in the spectrum of $[Ag_5Th_4]^{-1}$ is located at 463 nm (0.025), at 473 nm (0.037) and at 466 nm (0.061) according to RI-CC2, ADC(2), and M062X, respectively. The $[Ag_5Th_4]^{-1}$ spectrum is dominated by the transition at 438 nm (0.123), according to RI-CC2. This predominant transition is located at 423 nm (0.225) in the case of M062X. Once again CAM-B3LYP fails to predict the spectrum of the complex because its first transition is located at 416 nm (0.049).

Thus, ADC(2) can be used to compute the absorption spectra of silver-ligand complexes as an alternative to more computationally expensive methods, in particular coupled cluster singles and doubles. The usage of DFT is also possible, however, one should do it cautiously and with certain restrictions. CAM-B3LYP failed to reproduce the spectra for two out of six complexes, namely $[Ag_4Th_3]^{-1}$ and $[Ag_5Th_4]^{-1}$, but it can be recommended to calculate the absorption spectra of smaller complexes. M062X functional failed to reproduce the spectra of four out of six complexes, especially the spectra of $[Ag_2Th]^{-1}$ and $[Ag_2Th_2]^{-2}$.

Also, we compared the spectra calculated based on RI-CC2 and MP2 geometries (Table S7). The ADC(2) calculated spectra in both cases are almost identical, which once again demonstrates that MP2 optimization is an adequate alternative to a more computationally expensive CC2 method.

3.3 | DFT optimization and UV-Vis spectra

At last, we compared CC2 spectra with spectra calculated using DFT for both geometry optimization and calculation of the spectra with the same basis set def2-TZVP(P) (Table 3). CC2 and DFT spectra of $[Ag_2Th]^{-1}$ are consistent with each other. The CC2 spectrum of $[Ag_2Th_2]^{-2}$ starts at 580 and is dominated by the transition at 547 nm while DFT spectra start at approximately 440-480 nm with the predominant transition is located at 330-380 nm. The CC2 spectrum of $[Ag_3Th_2]^{-1}$ is pretty much consistent with B3LYP and CAM-B3LYP spectra while the maximum in the M06-2X spectrum is red-shifted by 50 nm. The dominant transition of $[Ag_3Th_2A]^{-1}$ in the CC2 spectrum is located at 413 nm while all DFT spectra are slightly shifted: to 431 nm, 393 nm, and 443 nm, according to B3LYP, CAM-B3LYP, and M06-2X. The same is true for $[Ag_4Th_3]^{-1}$: in the CC2 spectrum the predominant transition is located at 457 nm while all DFT maxima are hypsochromically shifted: to 420 nm, 378 nm, and 436 nm, according to B3LYP, CAM-B3LYP, and M06-2X. The DFT spectra of $[Ag_5Th_4]^{-1}$ more or less agree with CC2: the first transition has a low oscillator strength while the second is predominant located at 438 nm, 416 nm, 391 nm, and 434 nm, according to CC2, B3LYP, CAM-B3LYP, and M06-2X.

TABLE 3 First five excitation transitions (in nm; oscillator strengths—in parentheses) for silver-thiolate complexes optimized using DFT/def2-TZVP(P) approach

Method	$[Ag_2Th]^{-1}$	$[Ag_2Th_2]^{-2}$	$[Ag_3Th_2]^{-1}$	$[Ag_3Th_2A]^{-1a}$	$[Ag_4Th_3]^{-1}$	$[Ag_5Th_4]^{-1}$
B3LYP	422 (0.002)	454 (0.129)	402 (0.003)	529 (0.012)	532 (0.023)	424 (0.031)
	396 (0.554)	451 (0.133)	391 (0.589)	489 (0.007)	488 (0.004)	416 (0.139)
	347 (0.161)	364 (0.541)	338 (0.128)	431 (0.106)	429 (0.168)	378 (0.001)
	337 (0.006)	337 (0.013)	330 (0.003)	412 (0.082)	420 (0.224)	368 (0.000)
	334 (0.195)	337 (0.069)	324 (0.010)	390 (0.044)	391 (0.013)	365 (0.012)
CAM-B3LYP	393 (0.531)	441 (0.163)	389 (0.540)	393 (0.128)	486 (0.048)	392 (0.047)
	341 (0.229)	437 (0.167)	328 (0.217)	391 (0.005)	418 (0.215)	391 (0.173)
	336 (0.029)	335 (0.681)	318 (0.247)	361 (0.026)	384 (0.001)	324 (0.001)
	330 (0.234)	310 (0.029)	287 (0.011)	348 (0.000)	378 (0.327)	317 (0.000)
	295 (0.001)	307 (0.008)	275 (0.006)	336 (0.045)	345 (0.007)	312 (0.056)
M06-2X	412 (0.652)	486 (0.187)	435 (0.621)	443 (0.213)	557 (0.050)	444 (0.073)
	368 (0.223)	480 (0.190)	371 (0.185)	414 (0.004)	478 (0.202)	434 (0.230)
	357 (0.120)	387 (0.834)	362 (0.218)	386 (0.015)	436 (0.389)	362 (0.002)
	350 (0.157)	355 (0.006)	331 (0.005)	369 (0.100)	419 (0.018)	354 (0.334)
	299 (0.002)	349 (0.009)	290 (0.123)	353 (0.004)	382 (0.001)	321 (0.004)

Note: The spectra were calculated using the same DFT settings.
[a]A = O=CH_2; Th = S-CH_3.

To give an example, the computational cost of the one-thread calculation of the absorption spectra for the largest $[Ag_5Th_4]^{-1}$ complex using RI-CC2 method with def2-QZVP(P) split basis set is 522 h. The same calculation with ADC(2) method took 228 h. TDDFT calculations require less computational time: for example, M06-2X with def2-TZVP(P) basis set took only 2 h.

In conclusion, TDDFT spectra of silver-thiolates differ from CC2 to greater or lesser extent: in some cases the reason is different geometry (eg, B3LYP and $[Ag_3Th_2A]^{-1}$ complex) in other cases the reason is the character of the absorption spectra (eg, M06-2X and $[Ag_3Th_2]^{-1}$ complex).

4 | CONCLUSIONS

We compared the performance of different theoretical methods optimizing the geometry of silver-thiolate complexes. We showed that MP2 method gives reasonable geometries that are in good agreement with the geometries optimized at a higher level of theory, in particular, with CC2 method. DFT geometries differ from those obtained using CC2: in general, DFT tends to overestimate Ag—S and Ag—Ag bond lengths and fails to reproduce certain dihedral angles.

We compared the absorption spectra of silver-thiolate complexes simulated with different methods: CC2, ADC(2), and TDDFT. The spectra calculated with ADC(2) method were consistent with the spectra obtained with CC2. In several cases, TDDFT failed to reproduce the absorption spectra of silver-ligand complexes obtained at a higher level of theory. TDDFT spectra of silver-thiolates differ from CC2 to greater or lesser extent: in some cases the reason is different geometry (eg, B3LYP and $[Ag_3Th_2A]^{-1}$ complex) in other cases the reason is the character of the absorption spectra (eg, M06-2X and $[Ag_3Th_2]^{-1}$ complex while CAM-B3LYP fails to reproduce the spectrum of $[Ag_4Th_3]^{-1}$ complex). In our previous study, we have shown that M062X functional shows good accuracy as compared to ADC(2) ab initio method predicting the excitation spectra of silver NC complexes with nucleobases.[51] In this study we showed that M062X fails to reproduce the spectra of silver-thiolate complexes.

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The reported calculations were carried out on the Stallo supercomputer at the University of Tromsø.

AUTHOR CONTRIBUTIONS

A.A.B. prepared the original draft, visualized, and performed the investigation. V.A. P. devised the methodology, visualized, and performed the investigation. A.I.K. reviewed and edited the manuscript, supervised, and conceptualized.

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